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Integrating knowledge from knowledge graphs and large language models for explainable entity alignment

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ABSTRACT

Entity alignment, a critical task in integrating knowledge from multiple knowledge graphs (KGs), aims to identify equivalent entities across different KGs. Traditional approaches predominantly rely on knowledge embedding models to generate entity representations and compute similarity scores for alignment. However, these methods often lack interpretability, rendering their predictions opaque to end users. Recently, large language models (LLMs) have demonstrated strong semantic reasoning capabilities and have been applied to various KG-related tasks, including entity alignment. Despite this progress, existing methods still suffer from three key limitations: inaccurate retrieval of candidate entities, noisy prompt construction, and weak interaction between the retrieval module and the LLM. To address these challenges, we propose EARAG (entity alignment-oriented retrieval-augmented generation), a novel framework that effectively integrates structured knowledge from KGs with the semantic reasoning power of LLMs. EARAG first employs a convolutional neural network (CNN)-based retriever that jointly models multiple similarity metrics and captures relative ranking information to retrieve high-quality candidate entities. It then constructs carefully designed prompts that guide the LLM to not only determine entity equivalence but also generate human-understandable explanations. Extensive experiments on benchmark datasets demonstrate that EARAG achieves state-of-the-art alignment accuracy while offering superior interpretability. These results highlight the potential of retrieval-augmented LLMs as transparent and effective solutions for real-world entity alignment tasks. Code and datasets are publicly available at: https://github.com/linyaoyang/EARAG.

1. Introduction

Knowledge graphs (KGs) encapsulate real-world knowledge as triples, forming heterogeneous graphs where nodes denote concepts and edges represent the relationships between them [1]. In recent years, prominent KGs such as DBpedia [2] and Freebase [3] have experienced rapid development. Due to their inherently structured and interpretable nature, KGs offer a robust foundation for modeling knowledge and reasoning chains. As a result, they have been widely adopted in various knowledge-driven applications, including natural language understanding [4], smart diagnostics [5], and explainable artificial intelligence systems [6,7].

In practical applications, a KG is typically constructed from a single data source. However, due to the vastness and complexity of real-world knowledge, achieving comprehensive coverage within a single KG remains challenging. For instance, studies have shown that 71% of individuals in Freebase lack a recorded place of birth, and 75% have no information on nationality [8]. In contrast, different KGs often contain complementary or supplementary information. General-purpose KGs offer broad encyclopedic knowledge, whereas domain-specific KGs provide more specialized insights. Consequently, several studies advocate for integrating information from multiple KGs to fill knowledge gaps [9]. Entity alignment, which aims to identify equivalent entities across distinct KGs, plays a critical role in this integration process and has become a focal point of research in recent years.

Fig. 1 presents an example of the entity alignment process. This task typically involves computing similarities between entities across different KGs and identifying equivalent entity pairs based on these similarities. Early approaches to entity alignment were predominantly symbol-based, relying on string similarity between entity attributes, names, and descriptions [10]. These methods benefit from their use of explicit, human-interpretable attributes, which makes their alignment

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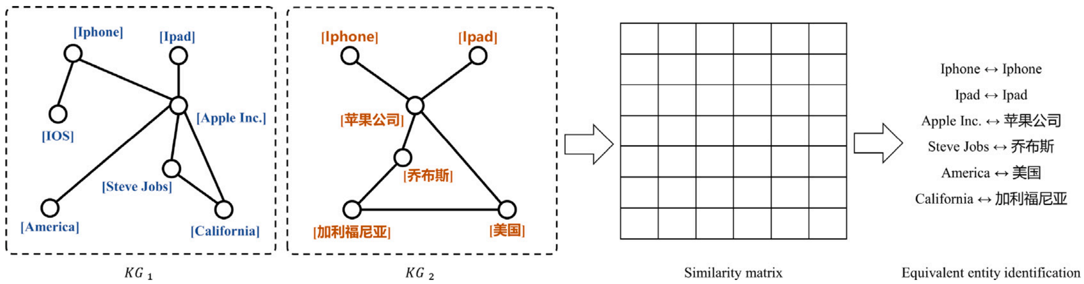

Fig. 1. Example of typical entity alignment process. Entity alignment methods typically calculate entity similarities across KGs using either symbol-based or embedding-based approaches to identify equivalent entity pairs.

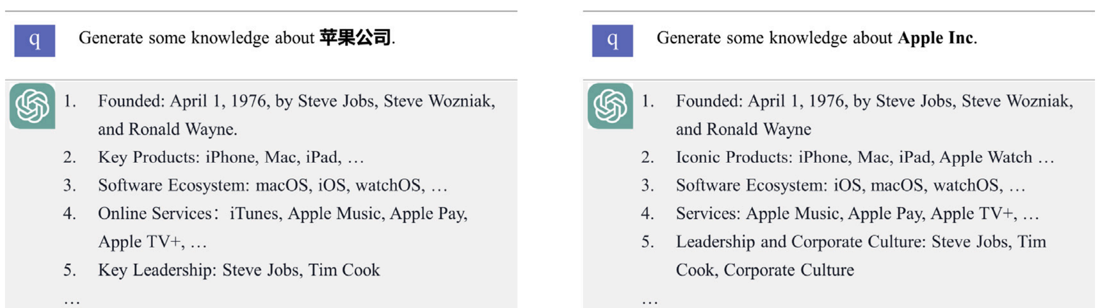

Fig. 2. Illustration of LLMs' capability to model semantic knowledge of entities across languages. Leveraging embedded knowledge, LLMs generate insights on entities with names in different languages, linking entities across linguistic and contextual boundaries with human-readable explanations for alignment.

decisions transparent and verifiable. However, symbol-based methods often struggle with cross-lingual KGs due to their reliance on exact or near-exact attribute matches. In contrast, more recent approaches leverage knowledge embedding models and graph neural networks (GNNs) [11–13] to generate dense vector representations that capture entity similarities. Some of these models focus on structural features – such as adjacency relationships, relation triples, and relation paths – to encode the structural context of entities. Others incorporate multiple types of information, including structure, name, and description, and integrate the resulting embeddings using various fusion strategies to compute overall similarity scores [14–16]. In general, embedding-based methods provide greater accuracy and flexibility than their symbol-based counterparts, particularly in complex or heterogeneous alignment scenarios.

Most existing entity alignment methods formulate the task of identifying equivalent entities as a ranking problem, wherein the candidate entity with the highest similarity score is selected. However, approaches that rely solely on embedding-based similarity face significant challenges in interpretability: while embeddings are capable of capturing various aspects of similarity, their high-dimensional and dense nature obscures the underlying factors contributing to similarity judgments. This opacity limits the practical utility of embedding-based alignment methods, especially in domains where transparency and accountability are critical. Furthermore, these methods are often constrained by the scarcity of labeled alignment data, which remains a significant bottleneck across many real-world datasets.

In recent years, large language models (LLMs) have achieved impressive performance across a broad range of natural language processing tasks. As model sizes have grown, modern LLMs have exhibited powerful emergent capabilities [17], enabling advances in areas such as code generation, recommendation systems, and gaming. Moreover, studies have shown that LLMs retain a substantial amount of factual knowledge embedded in their training data. As illustrated in Fig. 2, LLMs encode rich representations of real-world entities and implicitly capture semantic similarities between entities across KGs. Leveraging this internalized knowledge, LLMs can identify entities that refer to the same real-world object, even when they appear under different surface forms. Recent research has begun to explore the use of LLMs for entity alignment tasks [18–20], primarily by prompting the model to assess the equivalence of retrieved entity pairs. In these approaches, the LLM's output is either directly adopted as the final alignment decision or used to augment the training of embedding-based models. However, these methods often neglect the interpretability of the predictions—none explicitly utilize LLMs to explain the rationale behind their alignment decisions. In addition, existing LLM-based entity alignment approaches face several key challenges, including inaccurate retrieval of candidate entities, noisy prompt construction, and limited interaction between the retrieval module and the LLM.

To address these challenges, we propose a novel framework called entity alignment-oriented retrieval-augmented generation (EARAG), which integrates knowledge from both KGs and LLMs to identify equivalent entities and provide justifications for their alignment. The framework begins by computing comprehensive similarity scores between entities through a fusion of structural and name-based embeddings. Based on these scores, we retrieve the top-k candidate matches for each source entity. To further incorporate structural coherence, we identify a set of high-confidence equivalent entity pairs and calculate the number of shared equivalent neighbors for each candidate pair, thereby estimating neighborhood consistency. We then introduce a CNN-based retriever that jointly models structural similarity, name similarity, and neighborhood coherence. Crucially, the CNN is designed not only to capture these diverse signals but also to rank candidates by considering both their absolute similarity scores and their relative ranking among all candidates, leading to improved retrieval precision. In the subsequent generation stage, we construct a prompt template that incorporates entity names, associated triples, and similarity scores predicted by the CNN-based retriever. To minimize potential distractions from irrelevant or misleading context, we filter out non-equivalent neighbors from the prompt. This design encourages the LLM to focus on the core semantics of the entity pair and the attributes most relevant for determining equivalence. Extensive experiments demonstrate that our method effectively integrates structural, semantic, and neighborhood signals, achieving state-of-the-art performance on three widely used benchmark datasets. Moreover, analysis of the LLM-generated explanations shows that LLMs can produce interpretable alignment decisions, offering valuable insights into the rationale behind entity equivalence. The main contributions of this paper are summarized as follows:

- We propose a novel explainable entity alignment framework that integrates knowledge from both KGs and LLMs, enabling the retrieval of candidate entities and leveraging the extensive reasoning capabilities of LLMs to identify and justify entity equivalence.
- We introduce a new retrieval method that jointly models entities' structural features, name semantics, and structural coherence, while also incorporating relative ranking among candidates to improve retrieval accuracy.
- We design an innovative prompting strategy that allows LLMs to not only determine but also explain the equivalence between entities by incorporating entity names, associated triples, and predictions from the retriever. To minimize distractions and better guide the model's attention, we filter out non-equivalent neighboring entities, helping the LLM focus on core attributes relevant to alignment.
- We evaluate the proposed framework on three widely-used benchmark datasets, demonstrating that EARAG achieves strong performance and interpretability, underscoring the potential of LLMs to advance entity alignment.

2. Related work

Entity alignment, which aims to integrate multiple KGs by linking equivalent entities across them, has been the subject of extensive research in recent years. In this section, we provide a systematic review of traditional entity alignment methods, examining the diverse range of techniques developed in this field. Since our framework leverages LLMs to enhance entity alignment, we also present a concise overview of LLM applications in the context of KGs, with a particular focus on their recent progress in supporting entity alignment tasks.

2.1. Entity alignment

Existing entity alignment methods employ a variety of techniques to compute similarities between entities and identify equivalent pairs accordingly. These methods can be broadly classified into two categories: symbol-based approaches and embedding-based approaches.

Symbol-based entity alignment methods aim to link entities across KGs by comparing symbolic features such as string similarity of entity names, semantic similarity of descriptions, and structural relationships. Key techniques in this domain include string similarity measures like the Jaccard coefficient and Levenshtein distance, ontology-based reasoning, and rule-based systems that align entities according to predefined rules or heuristics. For example, PARIS [21] proposes a probabilistic framework for ontology alignment, leveraging probabilistic estimates of matching degrees derived from the functionality and equivalence of entities. Symbol-based approaches perform well when entity names and relationships are consistent and clearly defined across KGs, providing advantages in interpretability and transparency. However, these methods often struggle with incomplete or noisy data and face challenges arising from language variations and abbreviations in entity names. Moreover, they have limited capacity to capture deeper semantic meanings or contextual nuances, which may lead to missed alignments in complex or large-scale KGs.

In contrast, embedding-based entity alignment methods leverage representation learning techniques to obtain embeddings that capture entity similarities, thereby identifying equivalent entities based on the similarity of their embeddings. Some approaches focus exclusively on structural information from KGs, such as adjacency matrices, relational triples, and relation paths. Common structural embedding techniques include translation-based knowledge embeddings and GNNs. For example, MTransE [11] applies TransE [22] to independently learn embeddings for each KG and subsequently projects these embeddings into a shared vector space to enable direct similarity comparisons. GNNs, which more effectively capture global graph structures, aggregate information from neighboring entities to learn enriched representations [23]. GCN-Align [12], for instance, employs graph convolutional networks (GCNs) with shared parameters to embed entities within a unified vector space across multiple KGs. RDGCN [24] introduces a relation-aware dual-GCN that integrates relational information via attentive interactions between a KG and its dual relational graph. NAEA [25] further proposes a neighborhood-aware attentional representation method that exploits neighborhood subgraph-level information to improve embedding quality. Although these embedding-based approaches are scalable and flexible, they often overlook the rich semantic information contained in entities' attributes, which can constrain their effectiveness, particularly in sparser KGs.

To overcome these limitations, advanced embedding-based methods have been developed to capture diverse entity features. For example, AttrE [26] generates attribute embeddings from attribute triples, which are then used to align entity embeddings from two KGs into a shared vector space. DAT [27] employs a co-attention mechanism to dynamically weight structural and name embeddings, thereby enhancing similarity computation. BERT-INT [28] introduces a BERT-based interaction network to effectively fuse heterogeneous side information. Building upon this, MMIEA [14] extends BERT-INT to multimodal KGs by incorporating a multimodal interaction mechanism. DFMKE [16] proposes a multimodal fusion strategy that combines early and late fusion techniques: it first merges multimodal entity features via early fusion, followed by integration of heterogeneous knowledge through a low-rank multimodal late fusion approach, with the results combined using a dual fusion scheme. FuAlign [15] presents a cross-lingual entity alignment framework that leverages multi-view knowledge representations derived from a pre-fused KG.

Despite their strong performance, embedding-based methods exhibit limitations regarding their reliance on labeled equivalent pairs and a lack of interpretability. Most approaches formulate entity alignment as a ranking task, representing entities as embeddings but providing limited insight into the rationale behind the similarity judgments. Moreover, these methods often neglect the coherence of neighboring entities, which can serve as a crucial indicator of equivalence. Integrating both the quantity and similarity of aligned neighbors has the potential to enhance structural consistency and improve alignment accuracy—a factor that is frequently overlooked in existing approaches.

2.2. Large language models for knowledge graphs

LLMs are a class of language models pretrained on large corpora. Recently, with increases in model size, LLMs have demonstrated impressive capabilities, such as in-context learning and chain-of-thought reasoning, achieving notable success across various domains [17]. Recent research has also explored the potential of LLMs in KG-related tasks. For example, CodeKGC [29] utilizes the strong structural prediction and reasoning capabilities of LLMs to facilitate generative KG construction, enhancing knowledge extraction through schema-aware prompts and rationale-driven generation. StructGPT [30] introduces a unified framework to bolster LLM reasoning over structured data by developing specialized interfaces that collect relevant evidence, enabling LLMs to focus on reasoning based on the gathered information. KICGPT [31] presents a framework that integrates an LLM with a triple-based retriever for KG completion. ToG [32] proposes a responsible KG reasoning approach wherein the LLM acts as an agent that interactively explores related entities and relations within KGs, performing reasoning based on retrieved knowledge. By integrating LLM reasoning with expert feedback, ToG enhances both knowledge traceability and reasoning accuracy.

Recent studies have begun to explore the application of LLMs to enhance entity alignment. Some approaches aim to improve embedding-based alignment methods by enriching alignment seeds, leveraging

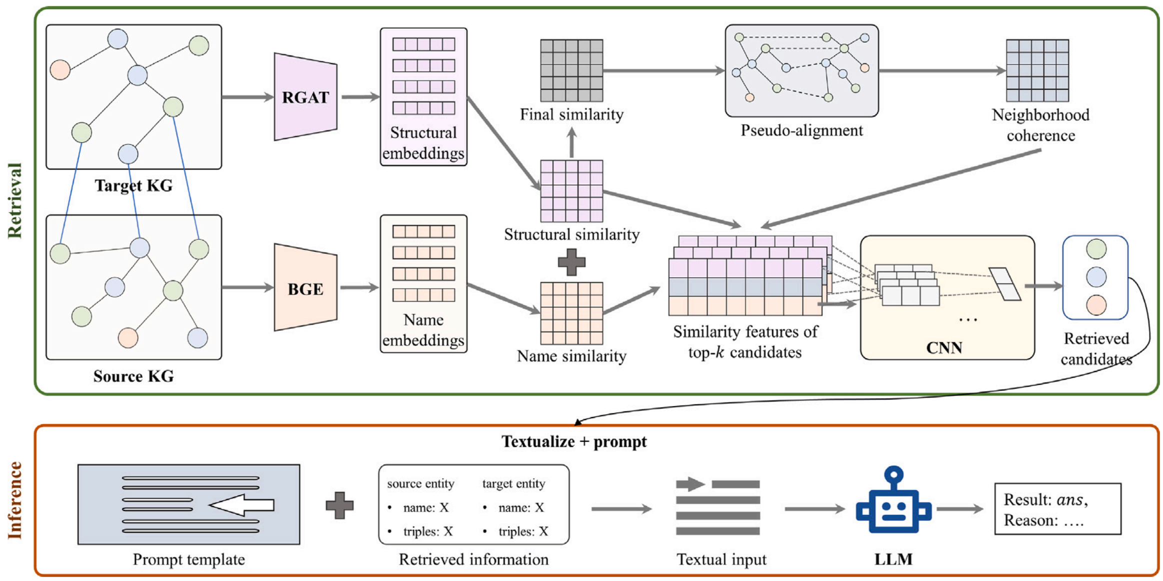

Fig. 3. Overall framework for EARAG. This framework adopts a retrieval-generation paradigm, comprising two main modules: the retrieval module and inference module. The retrieval module learns entity similarities considering structural and name embedding similarities and generates the top-k candidates for each source entity based on their similarities. It then pre-aligns high-confidence equivalent entities and counts the neighbor coherence similarities between entities, training a CNN-based retriever to retrieve the most possible candidate based on the structural, name, and neighbor coherence similarities. The inference module uses an LLM to determine and explain the equivalence between the source and retrieved entity.

the semantic reasoning capabilities of LLMs to generate high-quality pseudo-labeled pairs. For instance, LLM4EA [20] employs LLMs to generate pseudo-labels for entity alignment and integrates an active learning policy with an unsupervised label refinement mechanism to improve both performance and efficiency. AutoAlign [19] presents a fully automated approach that constructs predicate-proximity graphs using LLMs and aligns entities based on attribute similarity. Other studies directly prompt LLMs to assess entity equivalence, using their output as final predictions—often within a retrieval-then-prediction framework to reduce the computational cost of inference. For example, ChatEA [18] introduces a KG-to-code translation module that converts KG structures into textual representations, allowing LLMs to more effectively leverage background knowledge. Lippolis et al. [33] propose a hybrid framework that combines Wikidata entity search with a sequence-to-sequence LLM pipeline to align entities between Wikidata and ArtGraph. Seg-Align [34] categorizes samples into simple and hard cases, which are then handled by small language models (SLMs) and LLMs, respectively. HLMEA [35] reformulates the alignment task as a filtering and single-choice problem, integrating SLMs and LLMs to improve alignment performance without manual supervision.

Despite recent advancements, existing methods still exhibit several notable limitations. First, none of the current approaches utilize LLMs to provide explanations for their alignment predictions, leaving the interpretability of LLM-based entity alignment largely unexplored. Second, many methods suffer from suboptimal candidate retrieval, which directly impacts the quality of subsequent predictions. Third, prompt construction often introduces irrelevant or noisy contextual information, which can mislead the LLM and degrade performance. Finally, the interaction between the retriever and the LLM is typically loosely coupled, leaving room for further optimization.

To address these limitations, we propose EARAG, a novel framework that integrates a high-accuracy retriever for identifying candidate equivalent entities and employs a carefully designed prompting strategy that enables LLMs to both determine and justify entity equivalence.

3. Proposed model

This section begins with a formal definition of the entity alignment problem and then details the technical aspects of our proposed framework.

3.1. Preliminary

A KG can be represented as a tuple $G = (E, R, T)$, where $E = \{e_1, e_2, \dots, e_n\}$ denotes the set of entities, $R = \{r_1, r_2, \dots, r_m\}$ is the set of relations, and $T = \{(h, r, t) \mid h \in E, r \in R, t \in E\}$ represents the set of triples. Consider two related KGs, $G_1 = (E_1, R_1, T_1)$ and $G_2 = (E_2, R_2, T_2)$, that share some common entities. Let e denote a universal entity in the KGs, u represent an entity in G_1, and v represent an entity in G_2. Let $S = \{(u, v) \mid u = v, u \in E_1, v \in E_2\}$ be a seed set of labeled equivalent entity pairs. The task of entity alignment is to identify all equivalent entity pairs $C = \{(u, v) \mid u = v, u \in E_1, v \in E_2\}$ from the remaining set of entities. In the remainder of this paper, variables are represented in italics, sets in capital italics, vectors in bold, and matrices in bold capital letters.

3.2. Proposed framework

Recent state-of-the-art entity alignment methods typically identify equivalent entities based on embedding similarities. While effective in many scenarios, these approaches often overlook the underlying semantic relationships that underpin entity equivalence. Their reliance on purely numerical representations can obscure the true nature of equivalence, resulting in limited interpretability, as alignment decisions are primarily guided by similarity scores without offering insight into the underlying reasoning. Furthermore, most existing methods neglect to incorporate neighborhood coherence into their alignment process, despite its crucial importance in indicating entity equivalence.

To address these limitations, we propose EARAG, a framework that integrates knowledge from both KGs and LLMs to produce explainable entity alignment predictions. An overview of the proposed framework is shown in Fig. 3. Our approach follows a retrieval-generation paradigm, introducing a novel candidate retrieval module that jointly considers structural, name-based, and neighborhood similarities to identify the most probable candidate entities. The framework then prompts an LLM to both assess and justify the equivalence of the retrieved entity pairs, thereby enabling explainable alignment.

3.3. Candidate entity retrieval module

Given the substantial computational cost and interaction overhead of LLMs, it is impractical to evaluate every possible entity pair. Additionally, incorporating all target entities into an LLM prompt is challenging, as LLMs are constrained by finite context windows and are susceptible to truncation or omission of critical information—an issue commonly referred to as the lost in the middle phenomenon [36]. To address these challenges, incorporating a retrieval module is essential when applying LLMs to entity alignment tasks. While most existing entity alignment methods can serve as candidate retrieval models by selecting the most similar entities based on learned similarity matrices, they often neglect neighbor coherence, resulting in suboptimal retrieval performance.

The proposed candidate entity retrieval module integrates structural, name-based, and neighbor coherence similarities to identify the most probable candidate entities. Specifically, we first obtain entity name embeddings using a pretrained LLM. Subsequently, a relation-aware graph attention network (RGAT) is employed to learn structural embeddings, leveraging a seed set of equivalent entities augmented with high-confidence pairs derived from name embeddings. Name and structural similarities are then computed and fused via a weighted combination to produce a final similarity score between entities. Based on this similarity matrix, we perform pre-alignment of high-confidence entity pairs and compute the number of equivalent neighbors for each entity. Finally, for each source entity, the top-k candidate entities are selected, and a CNN-based retrieval module is trained to refine the selection of the final candidate entity.

3.3.1. Name embedding

Incorporating name information is critical for enhancing entity alignment performance, as entity names often carry key semantic cues indicative of identity. Prior approaches typically exploit name information in two primary ways. The first involves using pretrained word embeddings to represent entity names as the weighted average of the embeddings of their constituent words [15]. The second employs character-level similarity metrics, such as edit distance, to evaluate name similarity [37]. However, cross-lingual entity alignment poses additional challenges, as equivalent entities may have different names across languages. To mitigate this issue, existing methods often translate entity names into English to enable meaningful comparison.

Following this approach, we utilize the Google Translate API to convert non-English entity names into English. To improve the quality of name embeddings, we fine-tune a pretrained word embedding model to make it more domain-specific and better equipped to capture the nuances of entity names. In contrast to prior methods that represent names by averaging word-level embeddings, we treat each entity name as a complete sentence and generate a unified sentence-level embedding. This approach preserves the contextual relationships between words, capturing richer semantic information that is often lost in word-level representations. By leveraging sentence-level embeddings, our method can more effectively distinguish between entities with similar but non-identical names.

More specifically, we first translate all non-English entity names into English, denoting the translated name of an entity e as e_n. Entity pairs from the seed set are treated as positive samples, with their translated names considered equivalent. Using the BAAI general embedding (BGE) model pretrained on English text, we compute initial embeddings for all entity names:

$$\mathbf{e_n} = BGE\left(e_n\right). \tag{1}$$

To construct challenging negative samples, we select the top-15 most similar but non-equivalent entities for each positive pair, forming the negative set Q. This selection ensures that negative samples are semantically close to the positives, encouraging the model to capture subtle differences between similar entity names. Fine-tuning is performed using the InfoNCE loss, which optimizes the BGE model to maximize similarity for positive pairs while minimizing it for negative pairs:

$$\mathcal{L}_{bge} = -\sum_{(u_n,v_n)} \log \frac{e^{\langle \mathbf{u}_n,\mathbf{v}_n\rangle/\tau}}{e^{\langle \mathbf{u}_n,\mathbf{v}_n\rangle/\tau} + \sum_{(\tilde{u},\tilde{v})\in Q} e^{\langle \tilde{\mathbf{u}}_n,\tilde{\mathbf{v}}_n\rangle/\tau}}, \tag{2}$$

where $\langle\cdot,\cdot\rangle$ denotes the inner product of vectors, and τ is a temperature hyperparameter. After fine-tuning, the parameters of the BGE model are frozen, and refined name embeddings are generated as in Eq. (1).

Additionally, we experimented with Levenshtein edit distance for computing name similarity but found that it degraded candidate entity retrieval performance.

3.3.2. Structural embedding

Entities with similar names may represent entirely different concepts or roles within their respective KGs. For example, “Apple Inc.” and “Apple” may refer to distinct entities despite the lexical similarity. Structural information is crucial for resolving such ambiguities, as it provides contextual cues that help differentiate entities based on their surrounding relations. GNNs are particularly well suited for leveraging this structural context in entity alignment tasks. By propagating information through neighboring nodes, GNNs effectively capture both local and global structural patterns, enabling a more nuanced understanding of the relationships between entities.

To incorporate relational information during the aggregation of messages from neighboring entities, we adopt the RGAT model to learn structural embeddings. RGAT builds upon the Graph Attention Network (GAT) framework [23], a neural architecture specifically designed for structured data, which dynamically adjusts the importance of neighboring nodes through an attention mechanism. In a GAT layer, the hidden representation $\boldsymbol{h}_i \in \mathbb{R}^d$ of an entity e_i is computed as:

$$\boldsymbol{h}_i = \sigma\left(\sum_{j\in\mathcal{N}_i} \alpha_{ij}\mathbf{W}\boldsymbol{h}_j\right), \tag{3}$$

where d is the dimension of the hidden representation, $\boldsymbol{h}_j$ is the embedding of neighbor e_j generated by the current layer, and $\mathcal{N}_i$ denotes the set of neighbors of e_i. The matrix $\mathbf{W}$ is a learnable linear transformation, and the attention coefficient α_{ij} between e_i and e_j is calculated using a self-attention mechanism:

$$\alpha_{ij} = \frac{\exp\left(\eta\left(\mathbf{a}^T\left[\mathbf{W}\boldsymbol{h}_i \oplus \mathbf{W}\boldsymbol{h}_j\right]\right)\right)}{\sum_{u\in\mathcal{N}_i}\exp\left(\eta\left(\mathbf{a}^T\left[\mathbf{W}\boldsymbol{h}_i \oplus \mathbf{W}\boldsymbol{h}_u\right]\right)\right)}, \tag{4}$$

where $\mathbf{a} \in \mathbb{R}^{2d}$ is a learnable attention vector, $\oplus$ denotes vector concatenation, and η is the LeakyReLU activation function. In prior GAT-based entity alignment methods [38], the transformation matrix $\mathbf{W}$ is often constrained to a diagonal form to reduce the number of trainable parameters, mitigating the risk of overfitting and maintaining model generalization performance.

However, the vanilla GAT fails to capture the diversity of relationships between entities, thereby neglecting the relational proximity inherent in KGs. To address this limitation, several relation-aware entity alignment methods [39–41] incorporate relational information into the attention mechanism to better reflect the semantic connections between entities. Inspired by these approaches, we adopt an

Prompt template
You are an expert assistant. Analyze the entities from two different knowledge graphs and determine if they refer to the same real-world object based on their names, triples, and your own knowledge. Respond in the exact format: "result: Yes, reason: " or "result: No, reason: " depending on whether they represent the same object. Your explanation should only include essential information, with no line breaks or extra spaces. Be concise and accurate in your reasoning.
entity_1: {name: {name1}, triples: {triple1}}
entity_2: {name: {name2}, triples: {triple2}}
Note that the estimated similarity between these two entities is: {pred_logit (entity_1, entity_2)}

Fig. 4. Prompt template of EARAG. The template incorporates the entity names, selected triples, and similarity scores estimated by the CNN retriever into a prompt that instructs the LLM to determine whether the entity pair is equivalent.

RGAT to learn enriched entity embeddings. Specifically, the hidden representation output from an RGAT layer is computed as:

$$\boldsymbol{h}_i = \sigma\left(\sum_{j\in\mathcal{N}_i} \alpha_{ij}\mathbf{M}_{ij}\boldsymbol{h}_j\right), \tag{5}$$

where $\mathbf{M}_{ij}$ is a relation-aware transformation matrix determined by the relation between entities e_i and e_j. To project entity embeddings onto distinct relational hyperplanes and generate relation-specific representations, $\mathbf{M}_{ij}$ is defined as:

$$\mathbf{M}_{ij} = \boldsymbol{I} - 2\boldsymbol{h}_r\boldsymbol{h}_r^T, \tag{6}$$

where $\boldsymbol{I}$ denotes the identity matrix and $\boldsymbol{h}_r$ is the normalized embedding of the relation connecting e_i and e_j, ensuring $|\boldsymbol{h}_r|_2 = 1$. This formulation guarantees that $\mathbf{M}_{ij}$ is an orthogonal matrix:

$$\begin{aligned}\mathbf{M}_{ij}^T\mathbf{M}_{ij} &= \left(\boldsymbol{I} - 2\boldsymbol{h}_r\boldsymbol{h}_r^T\right)^T\left(\boldsymbol{I} - 2\boldsymbol{h}_r\boldsymbol{h}_r^T\right)\\ &= \boldsymbol{I} - 4\boldsymbol{h}_r\boldsymbol{h}_r^T + 4\boldsymbol{h}_r\boldsymbol{h}_r^T\boldsymbol{h}_r\boldsymbol{h}_r^T = \boldsymbol{I}.\end{aligned} \tag{7}$$

The orthogonality of $\mathbf{M}_{ij}$ ensures that when two entities are transformed within the same relational hyperplane, their norms and relative distances remain unchanged after transformation.

Similar to the standard GAT, α_{ij} denotes the attention coefficient between entities e_i and e_j, and is computed as follows:

$$\alpha_{ij} = \frac{exp\left(\boldsymbol{a}^T\boldsymbol{h}_{ij}\right)}{\sum_{u\in\mathcal{N}_i} exp\left(\boldsymbol{a}^T\boldsymbol{h}_{iu}\right)}, \tag{8}$$

where $\boldsymbol{h}_{ij}$ denotes the complementary edge embedding, defined as:

$$\boldsymbol{h}_{ij} = \sigma\left(\boldsymbol{h}_i + \boldsymbol{h}_j\right). \tag{9}$$

Since RGAT can only capture information from immediate neighbors, we stack multiple RGAT layers to aggregate multi-hop neighborhood information. The final structural embedding of entity e_i is obtained by concatenating the embeddings from all RGAT layers:

$$\boldsymbol{h}_i^{\text{out}} = \left[\boldsymbol{h}_i^0\left\|\boldsymbol{h}_i^1\right\|\cdots\|\,\boldsymbol{h}_i^l\right], \tag{10}$$

where $\boldsymbol{h}_i^0$ denotes the initial embedding of e_i, which is randomly initialized to provide diverse starting points for learning structural representations. $\boldsymbol{h}_i^l$ refers to the hidden representation of the lth RGAT layer.

The objective of the structural embedding model is to map equivalent entities closer together within a unified structural embedding space. To achieve this, we employ a pair of RGAT models with shared parameters to project entities from both KGs into a common vector space. The models are trained using a triplet loss function defined as:

$$\mathcal{L}_s = \sum_{(u,v)\in S}\left[\delta + dist\,(u,v) - dist\,(\tilde{u},\tilde{v})\right]_+, \tag{11}$$

where δ is a fixed margin, and $[x]_+ = \max(x, 0)$. The distance between the embeddings of entities e_i and e_j is computed as:

$$dist\left(e_i, e_j\right) = \left\|\boldsymbol{h}_i^{\text{out}} - \boldsymbol{h}_j^{\text{out}}\right\|_2^2. \tag{12}$$

Here, $\tilde{u}$ and $\tilde{v}$ are negative samples corresponding to u and v. To improve the model's ability to distinguish challenging entity pairs, we construct a negative sample pool that maintains the nearest non-aligned entity for each target in the vector space after each training step. This dynamic pool enables efficient retrieval of hard negatives, thereby enhancing both the model's performance and convergence speed.

Obtaining high-quality alignment seeds is challenging due to the limited availability of reliable equivalent entity pairs in real-world scenarios, which can impede the effective training of the RGAT model. To mitigate this issue, we expand the seed set by incorporating entity pairs that exhibit the highest similarity based on their name embeddings. This augmentation enhances the diversity of the alignment seeds and provides the model with a broader training set, thereby improving its learning capacity.

3.3.3. Neighbor coherence

Neighbor coherence is a valuable feature for entity alignment, as it captures the structural relevance of an entity's surrounding context. Specifically, it measures the number of equivalent neighbors shared by two entities across different KGs. A higher count of shared equivalent neighbors indicates a greater likelihood that the entities refer to the same real-world object, reflecting consistency in their relational structures. This structural signal is particularly valuable in cases where entity names are ambiguous, providing an additional layer of validation beyond name similarity. By integrating neighbor coherence, the alignment method becomes more robust and accurate, especially in scenarios where name-based techniques alone struggle to distinguish entities with similar surface forms but different underlying meanings.

Technically, we begin by computing both the structural and name embedding similarities between entities from G_1 and G_2. To measure the distance between embedding vectors, we employ the Bray-Curtis dissimilarity, defined as:

$$d\,(\mathbf{u},\mathbf{v}) = \frac{\sum_i\left(\left|u_i - v_i\right|\right)}{\sum_i\left(\left|u_i + v_i\right|\right)}, \tag{13}$$

where u_i and v_i represent the ith components of the embedding vectors $\mathbf{u}$ and $\mathbf{v}$. The Bray-Curtis dissimilarity normalizes the differences across embedding dimensions, thereby reducing the influence of scale variation. We then define the similarity score between $\mathbf{u}$ and $\mathbf{v}$ as $1 - d\,(\mathbf{u},\mathbf{v})$.

After computing the name embedding similarity matrix Sim_n and the structural embedding similarity matrix Sim_s, we combine them into a final similarity matrix by weighted averaging:

$$Sim = \lambda_1 Sim_s + \lambda_2 Sim_n, \tag{14}$$

where λ_1 and λ_2 are weighting hyperparameters. Based on this final similarity matrix, we identify pseudo-aligned entity pairs by selecting those that are mutually the most similar to each other in both directions. Specifically, if $Sim(u,v) = \max\,(Sim\,(u,:))$ and $Sim(u,v) = \max\,(Sim\,(:,v))$, then entities u and v are selected as a pseudo-aligned entity pair. For each candidate entity pair, we retrieve their matched neighboring entities and count the number of pseudo-aligned neighbors, denoted as Sim_c. This count serves as a neighbor coherence feature, which further aids in candidate entity retrieval.

Algorithm 1: Running process of EARAG

Input: KGs G_1 and G_2 with a few labeled entity pairs S
Output: Equivalent entity pairs with explanations
Generate name embeddings using a pretrained LLM and learn structural embeddings based on the seed set S;
Calculate final similarities between entities based on their embeddings;
Pre-align entities based on the final similarities;
Compute the neighbor coherence between entities;
Train the CNN-based retriever using the structural, name, and neighbor coherence similarities;
for e_i *in* E_1 **do**
    $t = 0$;
    while $t < \gamma$ **do**
        Generate the top-k candidate set C_i according to their final similarities to e_i;
        Predict the candidate equivalent entity e_j based on the similarity features;
        Filter out irrelevant triples based on the pre-alignment results;
        Construct prompt based on the name, relevant triples, and estimated similarities of e_i and e_j;
        Input the prompt into the LLM;
        if *The result is Yes* **then**
            Predict e_i and e_j as equivalent entities and extract explanations from the output reason;
            Break;
        else
            $t = t + 1$;
            Remove e_j from the target entity set;

3.3.4. CNN-based retriever

To reduce the computational overhead of LLM inference, we introduce a CNN-based retriever to integrate multiple heterogeneous features for efficient entity selection. Specifically, for each source entity, we first retrieve its top-k most similar candidate entities based on a combination of structural similarity, name semantic similarity, and neighbor coherence. These similarity values are organized into a $3 \times k$ feature matrix, where each row represents one type of similarity and each column corresponds to a candidate entity. The matrix is then reshaped into a tensor of shape (1, 3, k), with the similarity types treated as channels and the candidate entities as spatial features.

The CNN architecture processes the input through two convolutional layers followed by a fully connected layer. The first convolutional layer employs 16 filters of size 3×1 with a stride of 1 and no padding, effectively compressing the feature height – i.e., aggregating the three types of similarity – while preserving the width k. This results in an output tensor of shape (16, 1, k), which captures channel-wise interactions among the different similarity measures. The second convolutional layer applies 32 filters of size 1×1, serving as a feature transformation layer that increases the representational capacity along the channel dimension without altering the spatial resolution. The output of this layer is a tensor of shape (32, 1, k).

The resulting tensor is then flattened into a one-dimensional vector of size $32 \times k$ and fed into a fully connected layer, which maps it to a k-dimensional output vector. Each element in this vector corresponds to the matching score of a candidate entity. These scores are subsequently transformed into a probability distribution over the k candidates using the softmax function.

An important motivation for adopting a CNN in our retriever design is its inherent capability to model spatial dependencies [42]. This enables the model not only to capture diverse types of similarity but also to leverage the positional information of candidates within the retrieved list. Such positional cues are critical, as the relative ranking of candidates often carries valuable signals—e.g., higher-ranked entities are more likely to be correct matches. By applying convolutions along the candidate dimension, the CNN can learn local patterns that incorporate both similarity values and positional context, thereby enhancing retrieval accuracy.

During training, the model is supervised using labeled triples, where one candidate corresponds to the ground-truth aligned entity and the remaining $k-1$ candidates are hard negatives selected based on similarity ranking. To mitigate feature dominance and enhance training stability, the neighbor coherence scores are normalized by dividing each score by the maximum score within the candidate set for the given source entity. The model is trained using a standard cross-entropy loss:

$$\mathcal{L}_c = \sum_{i=1}^{|S|} \sum_{c=1}^{k} y_{i,c} \log\left(p_{i,c}\right), \tag{15}$$

where $y_{i,c}$ is a one-hot vector indicating the correct candidate for source entity e_i, and $p_{i,c}$ denotes the predicted probability of selecting the cth candidate.

At inference time, the CNN-based retriever efficiently filters the top-k candidates to identify the most probable match. This selected entity is then passed to the LLM for deeper semantic verification or explanation, thereby significantly reducing the number of LLM invocations and the overall computational cost. By learning local patterns across different similarity types and candidate entities, the CNN retriever acts as a lightweight yet effective pre-selection mechanism that achieves a balance between accuracy and efficiency.

3.4. Prompt template design

Traditional entity alignment methods primarily identify equivalent entities by selecting the candidate with the highest similarity to the source entity. However, these approaches often overlook the semantic context in which entities exist, making it difficult to justify why a particular match is made. By ignoring broader relational structures and contextual information, such methods lack transparent reasoning, ultimately compromising interpretability and trustworthiness. To overcome these limitations, we propose leveraging an LLM to assess the equivalence between source and candidate entities, while simultaneously generating natural language explanations to enhance interpretability and increase user confidence in the results.

To enable the LLM to make informed and interpretable alignment decisions, we design a structured prompt template that incorporates entity names, their associated triples, and similarity scores estimated by the CNN-based retriever, as shown in Fig. 4. Specifically, the softmax output probabilities from the CNN retriever are used as similarity scores for entity pairs. This template provides the LLM with both lexical and relational semantic information, facilitating context-aware

Table 1
The statistics of the evaluation datasets.

Datasets	KGs	Entities	Relations	Triples
$DBP15K_{ZH-EN}$	Chinese	19 388	1701	70 414
	English	19 572	1323	95 142
$DBP15K_{JA-EN}$	Japanese	19 814	1299	77 214
	English	19 780	1153	93 484
$DBP15K_{FR-EN}$	French	19 661	903	105 998
	English	19 993	1208	115 722
$SRPRS_{EN-FR}$	English	15 000	221	36 508
	French	15 000	177	33 532
$SRPRS_{EN-DE}$	English	15 000	222	38 363
	German	15 000	120	37 377
DY − NB	DBpedia	58 858	211	87 676
	Yago	60 228	91	66 546

and reasoning-driven equivalence judgments. The design mimics a semantic-informed reasoning process, where the LLM is first presented with background context before being tasked with assessing entity equivalence.

To reduce the noise caused by irrelevant or excessive triples – which we found to impair LLM performance during preliminary experiments – we implement a triple filtering strategy. Specifically, we retain only those triples whose neighboring entities have been identified as likely aligned pairs in the prior retrieval stage. For instance, if “New York City” and “NYC” are pre-aligned neighbors of the source and candidate entities, then triples such as (United States, has capital, New York City) and (USA, capital city, NYC) are preserved. This filtering reduces the input length and cognitive burden on the model, while focusing its attention on structurally significant and semantically meaningful overlaps.

By concentrating on this filtered, contextually relevant information, the LLM is better positioned to detect subtle yet significant similarities and differences between entities – such as shared roles, affiliations, or attributes – that may not be evident from names alone. Each prompt concludes with a clear instruction for the model to provide a binary equivalence decision (Yes/No), followed by a natural language explanation. This approach encourages the generation of interpretable justifications that can be reviewed by users or auditors.

The proposed prompt template provides several key advantages. First, it improves the prediction accuracy of LLMs by minimizing the impact of noisy or irrelevant triples. Second, it enhances interpretability by concentrating on triples that are more likely to represent meaningful relationships. Finally, it boosts computational efficiency by reducing the input size, which is especially important when performing entity alignment for large KGs.

The implementation of EARAG is detailed in Algorithm 1, comprising two main stages: retrieval and inference. In the retrieval stage, we first learn structural and name embeddings for entities from both KGs. Based on embedding similarities, a pre-alignment step is performed to identify pseudo-equivalent entity pairs. These pre-aligned pairs are then used to compute neighbor coherence similarities, which capture relational context that aids in identifying potentially equivalent entities. Leveraging structural, name, and neighbor coherence similarities as features, we train a CNN-based retriever to predict the most likely candidate entity for each source entity. During inference, for each unlabeled source entity, the retriever ranks the top-k candidate entities according to the computed features. The predicted similarity scores, together with the names and filtered triples of both the source and candidate entities, are passed to the LLM for evaluation. If the LLM confirms equivalence between the source and candidate entities, the alignment is accepted; otherwise, the candidate is discarded and the next top-ranked candidate is evaluated. This iterative process continues until an equivalence is confirmed or a predefined maximum number of iterations γ is reached. By integrating structural, lexical, and contextual information within a two-stage retrieval-then-inference pipeline, EARAG enhances both alignment accuracy and interpretability. The framework balances robustness and efficiency, providing a comprehensive solution for determining entity equivalence in complex, large-scale KGs.

4. Experiments

4.1. Experimental settings

To evaluate the effectiveness of the proposed framework and its components, we conducted experiments aimed at addressing the following five key research questions:

RQ1: Does our EARAG framework outperform state-of-the-art models in entity alignment tasks?

RQ2: How do the retrieval method and prompt template within our framework affect EARAG’s performance?

RQ3: Can LLMs effectively distinguish entity equivalence and provide reliable explanations for their predictions?

RQ4: How sensitive is EARAG to the selection of key hyperparameters?

RQ5: How efficient is EARAG in terms of LLM inference time and token consumption?

Datasets. We conduct experiments on three publicly available cross-lingual datasets: DBP15K [43], SRPRS [44], and DY-NB [45]. Table 1 summarizes the statistics of these datasets. DBP15K comprises three pairs of KGs extracted from DBpedia – ZH-EN, JA-EN, and FR-EN – linking entities in Chinese, Japanese, and French to their English counterparts. SRPRS, sampled from real-world KGs, preserves the natural distributions of entities and their relationships, along with reference links across different KGs. It contains two KG pairs, EN-FR and EN-DE, associating French and German entities with English ones. For our experiments, both DBP15K and SRPRS datasets are split into training and testing sets, with 30% of the equivalent entity pairs used for training and the remaining 70% for evaluation. DY-NB is a subset of the DWY-NB dataset, which also includes another subset named DW-NB. However, many entity names in DW-NB are anonymized into meaningless codes, lacking the semantic information necessary for effective inference by LLMs. Therefore, we exclude DW-NB from our evaluation. Following prior work [19], we evaluate and compare state-of-the-art baselines on the DY-NB dataset using varying proportions of seed alignments, ranging from 10% to 50%.

Parameter settings. For entity name embeddings, we employ the BGE small English model[1] to learn representations of entity names, with an output dimension of 384. The model is fine-tuned by generating 15 negative samples for each seed entity pair and is trained for 1000 epochs using the Adam optimizer with a learning rate of 1e−5. The temperature τ for the contrastive loss is set to 0.1. For structural embeddings, we implement a two-layer RGAT model, with each layer having a hidden dimension of 100. The model is trained for 10 epochs using the AdamW optimizer with a learning rate of 5e−3. The temperature δ for the structural loss function $\mathcal{L}_s$ is set to 8. To ensure stable training, we use mini-batches of size 1024 and apply cosine warm-up, early stopping, and gradient accumulation strategies. For the final similarity computation between entities, we set the weights λ_1 and λ_2 to 1. The CNN retriever is trained for 1000 epochs with a batch size of 1024, using the Adam optimizer and a learning rate of 1e−3. All key hyperparameters – including the number of filters, kernel sizes, and learning rate – are selected via grid search combined with 10-fold cross-validation on the training set to ensure optimal performance and generalizability. For entity equivalence inference, we use GPT-4 with the temperature set to 0.1 to reduce randomness. During inference, the maximum number of iterations γ is set to 5 to balance alignment accuracy and inference efficiency.

[1] https://huggingface.co/BAAI/bge-small-en-v1.5.

Table 2
Overall performance of different methods on different datasets.

Type	Model	DBP15K									SRPRS					
		ZH-EN			JA-EN			FR-EN			EN-DE			EN-FR		
		Acc	Hits@10	MRR	Acc	Hits@10	MRR	Acc	Hits@10	MRR	Acc	Hits@10	MRR	Acc	Hits@10	MRR
Structure-only	MTransE	30.83	61.41	0.36	27.86	57.45	0.35	24.41	55.55	0.34	10.70	24.80	0.16	21.30	44.70	0.29
	MuGNN	49.41	84.42	0.61	50.10	85.72	0.62	49.53	87.01	0.62	24.52	43.10	0.31	13.12	34.21	0.20
	AliNet	53.91	82.63	0.63	54.91	83.10	0.65	55.22	85.22	0.66	43.12	69.06	0.58	29.36	51.55	0.47
	RSN	58.02	81.17	0.66	57.41	79.86	0.65	61.24	84.14	0.69	50.02	72.34	0.59	34.57	60.88	0.44
	BootEA	62.94	84.75	0.70	62.23	85.39	0.71	65.30	87.44	0.73	50.31	73.20	0.58	36.52	64.92	0.46
	NAEA	65.01	86.73	0.72	64.14	87.27	0.72	67.32	89.43	0.75	30.70	53.50	0.39	17.70	41.60	0.26
	TransEdge	73.51	91.92	0.80	71.91	93.20	0.80	71.02	94.13	0.80	55.61	75.33	0.63	40.01	67.50	0.49
	DATTI	83.50	95.28	0.88	83.61	96.90	0.88	87.32	97.91	0.91	62.29	82.18	0.69	49.48	75.96	0.58
Multi-aspect	GCN-Align	41.25	74.38	0.54	39.91	74.46	0.54	37.29	74.49	0.53	38.50	60.01	0.48	24.29	52.17	0.41
	GM-Align	59.51	77.88	0.74	63.28	83.04	0.70	79.17	93.51	0.85	68.12	74.80	0.71	57.40	64.59	0.60
	RDGCN	70.75	84.55	0.75	76.74	89.54	0.81	88.54	95.62	0.89	77.89	88.62	0.82	67.20	76.69	0.71
	HGCN	72.01	85.67	0.75	76.61	89.70	0.81	89.19	96.08	0.90	76.30	86.29	0.80	67.01	76.98	0.71
	ESEA	72.47	86.94	0.78	74.99	87.78	0.79	79.39	90.88	0.83	71.12	84.61	0.74	63.92	74.49	0.67
	MRAEA	75.70	93.01	0.83	75.82	93.41	0.83	78.11	94.83	0.84	59.42	81.80	0.66	46.02	76.81	0.59
	CEA	78.01	–	–	85.07	–	–	93.41	–	–	86.30	–	–	78.22	–	–
	CEAFF	80.12	–	–	85.99	–	–	93.92	–	–	83.90	–	–	74.21	–	–
	RREA	82.19	95.62	0.96	91.81	97.23	0.98	96.28	98.99	**0.99**	80.14	91.01	0.83	75.33	84.89	0.78
	IPEA	85.71	–	–	90.28	–	–	97.31	–	–	88.22	–	–	79.97	–	–
	Dual-AMN	86.09	96.41	0.90	89.18	97.76	0.93	95.40	99.39	0.97	89.10	97.21	0.92	80.09	93.18	0.85
	RAGA	87.30	–	–	90.90	–	–	96.60	–	–	92.03	96.12	–	84.33	–	–
	PSR	88.28	98.18	0.93	90.78	98.68	0.94	95.80	99.88	0.97	88.08	96.98	0.91	80.78	93.26	0.85
	LLM4EA	94.86	98.67	0.96	96.46	98.85	0.97	96.87	99.18	0.98	93.10	98.70	0.95	80.20	96.00	0.86
	FuAlign	93.90	–	–	97.10	–	–	98.90	–	–	98.20	–	–	98.00	–	–
	ChatEA	95.18	96.54	0.97	97.25	99.14	0.98	99.00	**100.00**	**0.99**	–	–	–	–	–	–
	AutoAlign	–	–	–	40.20	67.50	0.50	55.20	79.30	0.64	–	–	–	–	–	–
	BERT-INT	96.76	98.99	0.97	96.39	99.10	0.97	**99.21**	99.79	**0.99**	–	–	–	–	–	–
	EARAG	**98.32**	**99.61**	**0.98**	**99.17**	**99.86**	**0.99**	99.03	99.95	**0.99**	**99.45**	**99.97**	**0.99**	**99.14**	**99.99**	**0.99**

Table 3
Performance of different methods on the DY-NB dataset.

Model	Seed: 10%			Seed: 20%			Seed: 30%			Seed: 40%			Seed: 50%		
	Acc	Hits@10	MRR	Acc	Hits@10	MRR	Acc	Hits@10	MRR	Acc	Hits@10	MRR	Acc	Hits@10	MRR
GCN-Align	7.41	23.04	0.12	15.47	41.79	0.24	22.52	53.49	0.32	28.37	61.80	0.38	32.44	65.27	0.42
RREA	14.23	31.61	0.22	23.14	55.19	0.33	31.11	65.99	0.42	34.31	69.83	0.46	38.73	74.65	0.51
CEA	99.71	99.86	0.99	99.71	99.86	0.99	99.71	99.86	0.99	99.71	99.86	0.99	99.71	99.86	0.99
CEAFF	96.32	96.46	0.96	96.32	96.46	0.96	96.32	96.46	0.96	96.32	96.46	0.96	96.32	96.46	0.96
AttrE	90.44	94.23	–	90.44	94.23	–	90.44	94.23	–	90.44	94.23	–	90.44	94.23	–
RDGCN	94.53	97.81	0.96	94.53	97.81	0.96	94.53	97.81	0.96	94.53	97.81	0.96	94.53	97.81	0.96
MRAEA	10.87	31.29	0.17	18.43	46.13	0.27	24.96	57.81	0.35	29.86	65.32	0.41	34.20	70.33	0.46
LLM4EA	70.10	85.33	0.76	70.10	85.33	0.76	70.10	85.33	0.76	70.10	85.33	0.76	70.10	85.33	0.76
FuAlign	99.97	–	–	99.98	–	–	99.98	–	–	99.99	–	–	99.99	–	–
AutoAlign	72.86	88.51	0.79	72.86	88.51	0.79	72.86	88.51	0.79	72.86	88.51	0.79	72.86	88.51	0.79
EARAG	**100**	**100**	**1**	**100**	**100**	**1**	**100**	**100**	**1**	**100**	**100**	**1**	**100**	**100**	**1**

Evaluation Metrics. Since the LLM's prediction of entity equivalence is framed as a binary classification task, we adopt accuracy as the primary evaluation metric, which measures the proportion of correctly identified equivalent entity pairs. To further assess the quality of the learned embeddings and the effectiveness of the retrieval component, we report Hits@10, which indicates the proportion of source entities whose true equivalent appears among the top 10 candidates ranked by similarity scores. This metric provides insight into how effectively the retrieval model reduces the candidate search space prior to LLM evaluation. In addition, we report the mean reciprocal rank (MRR) to evaluate the overall ranking quality of the true equivalent entities. MRR is calculated as the average of the reciprocal ranks of the correct matches and offers a more fine-grained assessment than Hits@10 by accounting for the exact position of the correct entity in the ranked list.

Baselines. We compare our proposed method against a comprehensive set of state-of-the-art entity alignment models. The selected baselines include both traditional embedding-based approaches and recent LLM-augmented methods:

- MTransE [11], which learns entity and relation embeddings for both KGs separately and then maps them into a unified vector space using a transformation matrix.
- BootEA [46], which enriches the training set with high-confidence alignment predictions, addressing limited labeled alignments.
- AliNet [13], which employs a novel GCN to aggregate information from both direct and indirect neighborhoods, helping manage non-isomorphic neighborhood structures.
- TransEdge [47], which introduces an edge-centric embedding model that contextualizes relation embeddings relative to specific head-tail entity pairs.
- MuGNN [48], which encodes KGs with multiple channels to generate alignment-oriented embeddings.
- RSN [44], which incorporates a recurrent skipping network to capture long-term dependencies based on relational path embeddings.
- NAEA [25], which integrates neighborhood subgraph-level information into entity embedding learning.
- MRAEA [40], which models cross-KG entity embeddings by attending to both incoming and outgoing neighbors and meta-semantic relations.
- RREA [39], which applies relational reflection transformations to produce relation-specific embeddings for each entity in a more efficient way.

- Dual-AMN [49], which proposes a dual attention matching network with a normalized hard sample mining loss to enhance alignment accuracy and improve speed.
- PSR [41], which proposes an entity alignment method optimized for high performance, scalability, and robustness without using negative sampling.
- RDGCN [24], which constructs dual-relation KGs and employs a dual-GCN to learn embeddings across both KGs.
- HGCN [50], which iteratively enhances entity and relation representations by approximating relation representations through entity embeddings.
- RAGA [51], which uses a relation-aware GAT to model interactions between entities and relations.
- GCN-Align [12], which captures both structural and attribute information of entities using a GCN.
- GM-Align [52], which introduces a topic entity graph for contextual entity information, using graph attention to align KG subgraphs.
- CEA [37], which frames entity matching as a stable matching task, based on a weighted combination of feature-based similarities.
- CEAFF [53], which utilizes a reinforcement learning framework to model entity interdependence and impose constraints on collective alignment.
- IPEA [54], which applies an integer programming-based algorithm to assign equivalent entity pairs based on comprehensive similarity measures.
- BERT-INT [28], which uses a BERT-based interaction model to align entities by evaluating neighboring entity interactions.
- ESEA [55], which combines symbolic and embedding approaches to address complex entity alignment tasks.
- DATTI [56], which enhances alignment through third-order tensor isomorphism, increasing decoding efficiency by leveraging adjacency and correlation isomorphisms within KGs.
- AttrE [26]: which integrates attribute-level character embeddings with structural embeddings for improved alignment.
- LLM4EA [20]: which leverages LLMs for automatic annotation and incorporates active learning and label refinement to improve alignment quality.
- FuAlign [15]: which enhances cross-lingual alignment by seed-based KG fusion and multi-view representation learning to capture semantic, contextual, and long-range information.
- ChatEA [18]: which integrates LLMs to leverage external knowledge and reasoning capabilities for improved alignment beyond traditional embedding-based methods.
- AutoAlign [19]: which enables fully automatic alignment via predicate-proximity graphs powered by LLMs and embedding-based matching using attribute similarities.

We exclude AttrE from the DBP15K and SRPRS datasets because it is only applicable in monolingual settings.

4.2. Performance analysis (RQ1)

In this experiment, we evaluate the effectiveness of EARAG in comparison to baseline models. The baselines are categorized into two groups based on the types of information used for entity representation learning: structure-only methods and multi-aspect information-based methods. Structure-only methods – such as MTransE, BootEA, and RSN – leverage the adjacency matrix and relational information to learn entity embeddings. In contrast, multi-aspect methods incorporate additional features, including entity names, attributes, and textual descriptions. Most baseline methods perform alignment by directly selecting the most similar candidates, although some approaches, such as CEA, IPEA, and CEAFF, apply additional mechanisms to enforce one-to-one alignments. To ensure a fair comparison, we report benchmark results separately for each group.

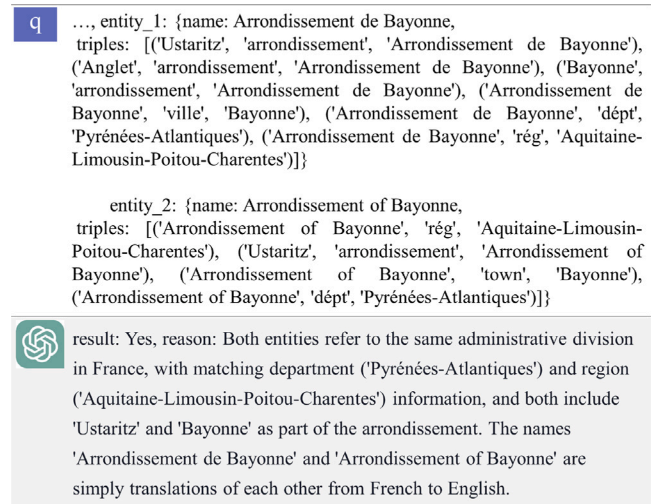

Fig. 5. An example of the LLM correctly identifies and explains the equivalence of two entities in EARAG.

Table 4
Accuracies of different ablation settings.

Settings	ZH-EN	JA-EN	FR-EN	Avg.
EARAG	98.32	99.17	99.03	98.84
w/o name embedding	94.61	98.17	97.65	96.81
w only BGE name embedding	77.56	89.39	91.72	86.22
w/o structural embedding	84.85	91.61	92.89	89.78
w/o CNN retriever	95.30	98.20	97.93	97.14
w/o LLM	**99.10**	**99.85**	**99.86**	**99.60**
w/o name	93.85	97.18	95.59	95.54
w/o triples	96.13	94.33	95.08	95.18
w full triples	96.42	98.88	98.68	97.99
w/o similarity	97.24	98.86	98.53	98.21

Table 2 presents the results on the DBP15K and SRPRS datasets, with the best-performing metrics across all models highlighted in bold. Some metrics are unavailable for certain methods due to differences in experimental settings and are indicated with a '–'. For instance, CEA, CEAFF, and IPEA enforce one-to-one alignment using stable matching, reinforcement learning, and integer programming, respectively, and therefore do not report Hits@10 on all datasets. Additionally, BERT-INT computes similarity based on entity descriptions, which are not available in the SRPRS dataset. The Hits@10 and MRR scores reported for our EARAG model are calculated from the final similarity matrix to assess the quality of the learned embeddings.

Structure-only vs. multi-aspect information. The results demonstrate that multi-aspect information-based methods consistently outperform structure-only approaches, underscoring the benefits of incorporating diverse types of information to capture more comprehensive entity similarities. Among structure-only methods, GCN-based models generally outperform those based on TransE, as they are more effective in capturing both local and global structural proximities and support end-to-end embedding learning. Furthermore, relation-aware embedding techniques such as RREA tend to yield superior performance compared to traditional structural embeddings, indicating the importance of relational information in enhancing graph representations. In addition, contextual features – such as entity names, attributes, and descriptions – provide crucial distinctions between structurally similar entities, further improving the effectiveness of multi-aspect approaches.

Our method vs. Baselines. The results indicate that EARAG consistently achieves the highest entity alignment accuracy across most datasets, demonstrating its effectiveness in accurately retrieving equivalent entities and enabling the LLM to reliably distinguish between equivalent and non-equivalent pairs. While BERT-INT slightly outperforms EARAG on the FR-EN dataset, it heavily relies on entity descriptions, which limits its scalability and renders it inapplicable to datasets like SRPRS where such information is unavailable. Moreover, similar to other baselines, BERT-INT lacks interpretability in its predictions. Although ESEA improves alignment performance by combining symbolic and embedding-based approaches, it remains less accurate than our method and does not explicitly address the issue of interpretability. Existing LLM-enhanced entity alignment methods also fall short in performance and overlook interpretability, further highlighting the strengths of our retrieval module and LLM-based inference mechanism. As illustrated in Fig. 5, EARAG not only correctly identifies equivalent entity pairs but also generates coherent explanations based on their underlying semantic information, demonstrating its strong interpretability.

We also report the results of several state-of-the-art methods on the DY-NB dataset in Table 3. As shown, our proposed EARAG method consistently outperforms all baselines across different seed ratios, achieving perfect scores in accuracy, Hits@10, and MRR. These results underscore the robustness and effectiveness of EARAG in aligning equivalent entities within the DY-NB dataset. A key contributor to this strong performance is the high degree of name similarity between equivalent entities in this dataset. This characteristic particularly favors methods that leverage high-quality name embeddings – such as FuAlign and RDGCN – as the semantic information encoded in entity names alone is often sufficient to accurately identify equivalent pairs.

4.3. Ablation study (RQ2)

To assess the effectiveness of the proposed modules within the EARAG framework, we conducted ablation studies on three KG pairs, with the results summarized in Table 4. We first evaluated EARAG's performance when the retrieval module operates without name embeddings, structural embeddings, or the CNN-based retriever. Excluding name embeddings from the similarity computation led to a noticeable drop in accuracy, underscoring the importance of name semantics in identifying equivalent candidate entities. Similarly, removing structural embeddings resulted in a substantial performance decline, with an average accuracy decrease of 9.06%, demonstrating the critical role of structural information in the retrieval process. Moreover, bypassing the CNN-based retriever – by selecting the top candidate solely based on similarity scores – caused a 1.7% drop in final prediction accuracy, highlighting the retriever's contribution to enhancing entity retrieval quality. We also evaluated alignment performance using only name embedding similarities-specifically, BGE embeddings-to assess their effectiveness in cross-lingual retrieval. The results show that relying solely on BGE embeddings fails to achieve optimal performance, particularly on the ZH-EN dataset, indicating that the additional components of our retrieval module are essential for effective cross-lingual entity alignment.

Next, we evaluate the performance of EARAG when the LLM-based entity alignment prediction is removed—i.e., retrieved candidate entities are directly assigned as equivalents to source entities, similar to traditional embedding-based methods. The results show a decrease in alignment accuracy when the LLM is employed to verify retrieved pairs. This drop is not attributable to the quality of the retrieval, which remains relatively high, but rather to the LLM occasionally misclassifying truly equivalent entity pairs as non-equivalent. Such misclassifications are likely due to hallucinations or a lack of factual grounding in the LLM, leading it to overemphasize superficial differences in entity names or associated triples. Nonetheless, removing the LLM also eliminates interpretability, as the system no longer provides reasoning behind alignment decisions. Our approach strikes a balance by integrating LLM-based reasoning to enhance interpretability while maintaining strong alignment performance.

Table 5
Accuracies of different LLM inference strategies.

Settings	ZH-EN	JA-EN	FR-EN
EARAG with Llama3.1-8B	**97.24**	**98.37**	**98.05**
Iterative inference with Llama3.1-8B	5.12	5.67	4.96

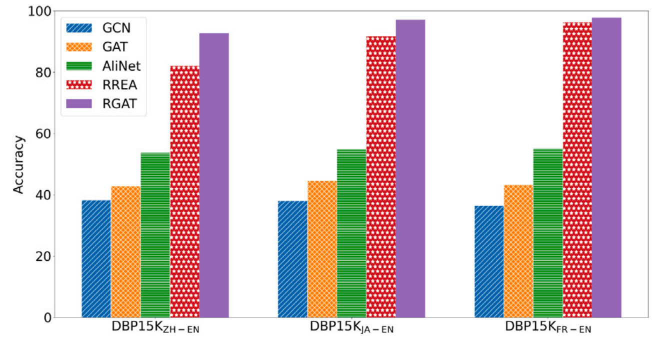

Fig. 6. Entity retrieval accuracy of different structural embedding models.

To evaluate the effectiveness of our retrieval-augmented alignment strategy, we perform an ablation study comparing EARAG with a baseline that conducts iterative inference using the same LLM (Llama3.1-8B) but without the support of a retriever. As presented in Table 5, EARAG significantly outperforms the baseline across all language pairs – ZH-EN (97.24%), JA-EN (98.37%), and FR-EN (98.05%) – whereas the iterative inference baseline achieves accuracies below 6%. This stark contrast underscores the dual advantages of the retriever module. First, it dramatically reduces inference costs by narrowing the search space to a small set of relevant candidate entities, thereby minimizing unnecessary LLM queries. Second, and more critically, it improves alignment accuracy by mitigating the hallucination tendencies of LLMs, which may incorrectly label superficially similar entities as equivalent. By filtering and presenting only the most semantically relevant candidates, the retriever steers the LLM toward more accurate equivalence judgments and reduces the impact of irrelevant or misleading entities.

To evaluate the impact of key components in our proposed prompt template, we conduct experiments comparing the performance of EARAG with its variants that modify the prompt by omitting entity names, removing filtered associated triples, including all associated triples, or excluding estimated similarity scores. The results reveal that omitting entity names from the prompt often causes the LLM to misidentify the target entity, resulting in decreased alignment accuracy. Similarly, removing filtered associated triples leads to an average accuracy drop of 3.66%, demonstrating that these triples provide valuable contextual cues for identifying equivalent entities, especially when entity names differ. In contrast, including all associated triples without filtering also degrades performance. This decline arises because irrelevant neighboring entities introduce noise, prompting the LLM to overemphasize structural discrepancies and mistakenly classify equivalent entities as inequivalent. These findings underscore the importance of incorporating both entity names and carefully filtered associated triples for optimal performance. Moreover, removing the similarity scores estimated by the CNN-based retriever leads to a further decline in accuracy – from 98.84% to 98.21% on average – highlighting the critical role of explicitly including similarity information in the prompt. These scores provide strong semantic signals that guide the LLM's inference, reinforcing its understanding of the closeness between candidate entity pairs.

To evaluate the effectiveness of the selected structural embedding model, we conduct experiments comparing its performance with that of several GNN-based methods, including GCN-Align, GAT, AliNet, and

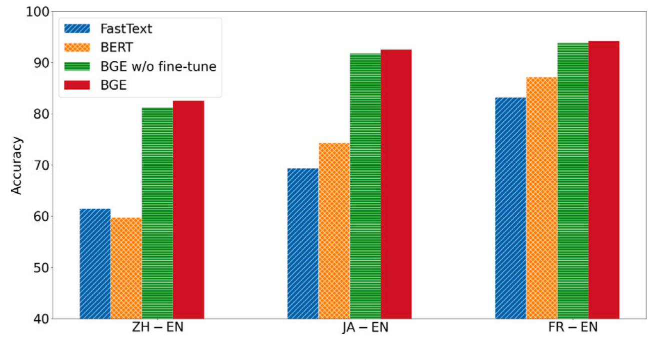

Fig. 7. Entity retrieval accuracy with different name embedding methods.

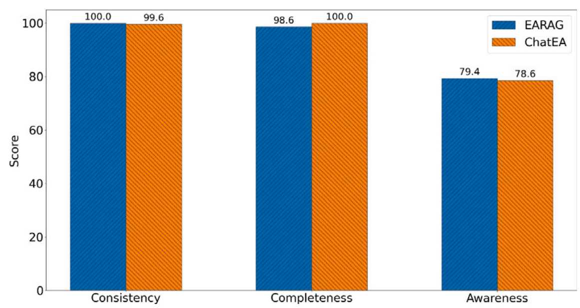

Fig. 8. Interpretability performance of the LLM.

Table 6
Entity alignment performance with different LLMs.

LLMs	ZH-EN	JA-EN	FR-EN	Average
GPT-4	**98.32**	**99.17**	**99.03**	**98.84**
GPT-3.5	98.02	98.83	98.46	98.44
Qwen-max	97.97	98.93	98.13	98.34
Llama-3.1-8B	97.24	98.37	98.05	97.89
Llama-3.1-70B	98.08	98.95	98.56	98.53

RREA. Specifically, we assess how well these structural embeddings capture the underlying structural similarities between entities by measuring retrieval accuracy based on embedding similarity, as shown in Fig. 6. The results demonstrate that methods incorporating relational information into the structural embedding process – such as RREA and RGAT – substantially outperform those that rely solely on adjacencies. Notably, RGAT achieves the highest retrieval accuracy among all baselines, highlighting its superior ability to capture structural similarities between entities.

We further evaluate the effectiveness of the name embedding model used in EARAG by comparing the BGE model against two widely adopted name embedding methods, as well as a non-fine-tuned variant of BGE. As shown in Fig. 7, BGE substantially outperforms both FastText and BERT in capturing the semantic similarities of entity names across languages. Additionally, fine-tuning the BGE model leads to further performance gains, demonstrating its enhanced capability to learn name embeddings specifically tailored for entity alignment.

4.4. Analysis of LLMs' predictions (RQ3)

LLMs exhibit varying levels of performance on knowledge-intensive tasks. To assess their effectiveness in entity alignment, we evaluate several state-of-the-art LLMs across three datasets, including both open-source and proprietary models such as GPT-3.5, Qwen-max, LLaMA 3.1-8B, and LLaMA 3.1-70B. As shown in Table 6, most models achieve high alignment accuracy, demonstrating a strong ability to identify equivalent entities. Notably, although the LLaMA 3.1-8B model lags slightly

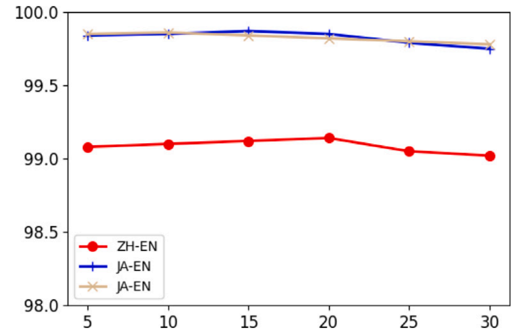

Fig. 9. The effect of k.

behind its larger counterparts, its average accuracy is only 0.95% lower than that of the best-performing model, GPT-4. These results highlight the potential of smaller LLMs to offer competitive performance in entity alignment while serving as a cost-effective alternative for further development.

4.5. Parameter sensitivity (RQ4)

To assess the accuracy of LLM-generated explanations for entity equivalence decisions, we randomly selected 1000 equivalent entity pairs and invited five PhD-level experts to manually evaluate the quality of each explanation. The evaluation was based on three criteria: consistency, completeness, and awareness. Consistency examines whether the explanation aligns with the LLM's final prediction—an explanation is deemed consistent if it logically supports the model's decision on whether the entities are equivalent. Completeness assesses the extent to which the explanation considers both entity names and their associated triples, reflecting how thoroughly the LLM utilizes contextual information. Awareness evaluates whether the LLM explicitly or implicitly identifies the real-world object that the entities represent, indicating its ability to ground reasoning in semantic understanding rather than relying solely on surface-level similarity.

Since most existing LLM-based entity alignment methods do not require the model to explain its predictions, direct comparisons with prior baselines are inherently limited. Nevertheless, we include a quantitative comparison with ChatEA, which incorporates partial reasoning by comparing entity names, attributes, and descriptions in its outputs. As illustrated in Fig. 8, the explanations generated by LLMs within our framework are generally consistent with their predictions, enhancing the overall reliability of the results. In the majority of cases, the LLMs compare entities using both name information and structural context, facilitating accurate alignment. However, in 20.6% of instances, the LLM relies solely on surface-level similarity, without explicitly referencing the real-world entity being represented. This observation reveals a limitation in the model's entity awareness and underscores the need for further improvements in semantic grounding.

Compared to ChatEA, our EARAG framework demonstrates higher performance in terms of consistency and awareness, though it slightly lags in completeness. This is largely because ChatEA explicitly prompts the LLM to assess entity similarity across multiple dimensions – such as names, structure, and descriptions – and to base its predictions on these similarity scores. While this design encourages comprehensive comparisons, it may inadvertently lead the model to prioritize numerical similarities over the recognition of real-world referents. As a result, inconsistencies may arise between the alignment decisions and the underlying semantic grounding of the entities.

In general, expanding the candidate set in the entity retrieval module increases the likelihood of retrieving the correct equivalent entity. To investigate the effect of the candidate count k on retrieval performance, we conducted experiments on three KG pairs while varying

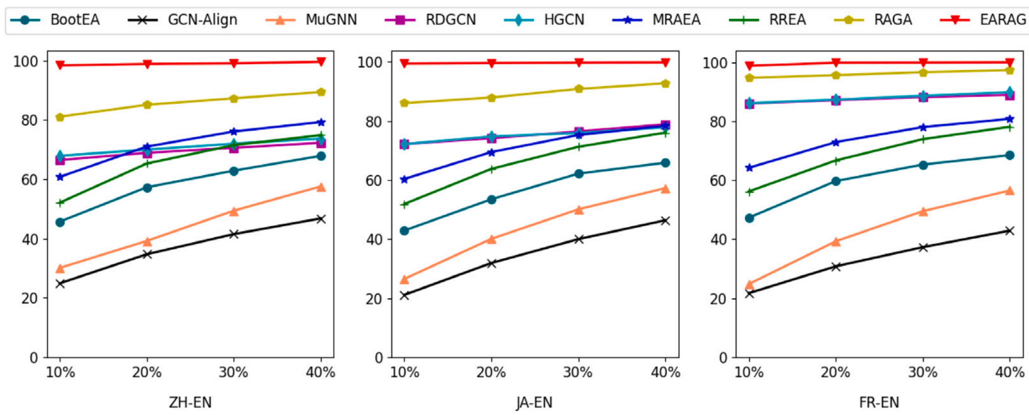

Fig. 10. Candidate entity retrieval performance with respect to the proportion of pre-alignment seeds.

Table 7
Efficiency analysis of EARAG with different LLMs. avg. tokens and avg. time (s) represent the average tokens and time (seconds) cost of EARAG per candidate entity.

LLMs	ZH-EN		EN-DE	
	Avg. tokens	Avg. time (s)	Avg. tokens	Avg. time (s)
GPT-3.5	63.34	1.62	57.48	1.66
GPT-4	92.82	14.61	72.07	10.92
Qwen-max	65.33	3.89	52.18	2.97
Llama3-8B	61.12	5.53	59.78	29.18
Llama3-70B	77.93	8.13	65.83	6.99

this parameter. The retrieval accuracies for different values of k are shown in Fig. 9. The results indicate that accuracy initially improves as the candidate set size increases, reaching a peak between 5 and 30 candidates, before gradually declining. This trend suggests that a larger candidate set enhances the probability of including the correct equivalent entity, thereby improving alignment accuracy—up to a point. However, further enlarging the candidate pool introduces more distractors, making the alignment task more difficult. Moreover, a larger candidate set increases the number of parameters in the retrieval model, which may lead to overfitting and ultimately reduce performance.

Following established practice, we conduct experiments on the DBP15K dataset using pre-aligned seed proportions ranging from 10% to 40% to assess the sensitivity of the entity retrieval model to the number of alignment seeds in cross-lingual KGs. Fig. 10 presents the performance of our entity retrieval module alongside several top-performing baseline methods. The results show a consistent improvement in performance across all methods as the number of pre-aligned seeds increases. Notably, EARAG outperforms all baselines across all experimental settings and datasets, highlighting its effectiveness. Moreover, our model exhibits greater robustness compared to most baselines in low-resource scenarios: while the accuracy of many methods drops significantly with fewer alignment seeds, EARAG maintains strong performance even with just 10% of alignments used for training. This robustness stems primarily from the use of seed-independent name information to enhance candidate generation, ensuring a high recall rate. In addition, the powerful candidate selection capabilities of our retrieval module enable accurate identification of equivalent entities within the candidate pool.

4.6. Efficiency analysis (RQ5)

The average number of tokens generated by each LLM during entity alignment tasks, along with the corresponding average inference time, is summarized in Table 7. Due to differences in the APIs used to access each model, response times do not necessarily correlate with model size. Specifically, the Llama models are accessed via Nvidia's APIs,[2] [3] while the Qwen-max model is accessed through Qwen's Dashscope API.[4] As expected, for any given LLM, generating a greater number of tokens results in longer reasoning times. The proposed prompt template facilitates more concise responses by guiding the model toward both accurate entity alignment predictions and succinct explanations, thereby reducing token output and improving inference efficiency. EARAG is particularly suited for application scenarios that demand high accuracy and interpretability in entity alignment, even at the cost of some computational overhead. Additionally, its flexible design enables seamless integration with different LLMs, allowing it to benefit from future advances in LLM inference speed and accuracy.

5. Conclusions

Entity alignment is crucial for enabling information fusion across heterogeneous KGs. While many existing methods achieve strong performance by learning entity embeddings from multiple information sources and identifying equivalent entities based on embedding similarity, they often lack interpretability. LLMs, with their vast knowledge and powerful reasoning capabilities, offer the potential to identify equivalent entities while also providing human-readable explanations. To support explainable entity alignment, we propose an entity alignment-oriented retrieval-augmented generation (EARAG) framework that integrates knowledge from both LLMs and KGs. EARAG combines a relation-aware graph attention network with a pretrained language model to jointly learn structural and name-based embeddings for entities. These embeddings are used to pre-align pseudo-equivalent entity pairs and retrieve the top-k most similar candidates. To further evaluate candidate quality, neighbor coherence similarity is computed to assess the alignment consistency of neighboring entities. A CNN-based retriever is then employed to select the most likely match from the top-k candidates. Finally, a prompt template incorporating entity names, associated triples, and estimated similarities is constructed to guide the LLM in determining entity equivalence and generating an explanation based on both input information and its internal knowledge. By integrating a lightweight retrieval model with LLM-based generation, our framework enhances both the accuracy and interpretability of entity alignment. Extensive experiments on three public datasets demonstrate that EARAG significantly outperforms state-of-the-art methods, highlighting the effectiveness of combining structured knowledge from KGs with the reasoning ability of LLMs.

Despite these promising results, the EARAG framework has several limitations. First, due to the hallucination problem of LLMs, generated

2 https://build.nvidia.com/meta/llama-3_1-8b-instruct.
3 https://build.nvidia.com/meta/llama-3_1-70b-instruct.
4 https://dashscope.console.aliyun.com/model.

predictions may sometimes contain inaccurate or fabricated content, which can compromise overall alignment accuracy. This highlights a fundamental trade-off between interpretability and accuracy: overreliance on LLM outputs may degrade precision, whereas minimizing LLM involvement may reduce transparency. Second, the quality of LLM predictions is highly sensitive to prompt design. In our current implementation, prompts are manually crafted and iteratively refined based on empirical observations. This approach poses scalability challenges and may limit generalizability across domains, while also introducing potential human biases.

In future work, we aim to address these limitations by developing more principled strategies for balancing accuracy and interpretability. One promising direction is the design of end-to-end retrieval-generation pipelines that jointly optimize retrieval quality and LLM reasoning. Another avenue involves enhancing the alignment capabilities of LLMs through supervised fine-tuning or reinforcement learning, using retrieval feedback to guide the learning of structural alignment patterns. Additionally, incorporating confidence calibration techniques and validating LLM outputs against external knowledge sources or logical constraints could help mitigate hallucinations and improve the reliability and trustworthiness of the system.

CRediT authorship contribution statement

Linyao Yang: Writing – original draft, Visualization, Software, Methodology, Funding acquisition, Conceptualization. **Hongyang Chen:** Writing – review & editing, Supervision, Funding acquisition. **Xiao Wang:** Writing – review & editing, Supervision, Conceptualization. **Jing Yang:** Writing – original draft, Software, Methodology. **Fei-Yue Wang:** Writing – review & editing, Supervision, Conceptualization. **Han Liu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Linyao Yang reports article publishing charges was provided by National Natural Science Foundation of China under Grant. Hongyang Chen reports article publishing charges and statistical analysis were provided by National Key Research and Development Program of China. Linyao Yang reports a relationship with Chinese Academy of Sciences Institute of Automation that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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